Voice Forgery Detection by Combining Spectro-Temporal Representations — AASIST

Jung et al. (2022), in their paper *AASIST: Audio Anti-Spoofing Using Integrated Spectro-Temporal Graph Attention Networks*, present a deep learning architecture for detecting voice forgeries, based on combining temporal and spectral information by means of Graph Attention Networks. The study focuses on the challenge of detecting artifacts indicating fake speech, which may appear in different time segments and frequency ranges depending on the method of creating the forgery. As an alternative to complex systems that combine several models, the researchers propose a single End-to-End system capable of learning relations between the different types of information.

The architecture receives a raw audio signal (Raw Waveform) and uses an Encoder based on RawNet2 to extract initial features. These features are used to build two graph representations, temporal and spectral, which are combined by means of a Heterogeneous Stacking Graph Attention Layer (HS-GAL). This layer includes an Attention mechanism that distinguishes between relations within each of the graphs and relations between the graphs, as well as a Stack Node that serves to accumulate combined information. In addition, the researchers present a Max Graph Operation (MGO) mechanism, based on two parallel branches and a Maximum operation between their outputs, with the aim of encouraging the learning of complementary features. A dedicated Readout mechanism combines the information accumulated in the graph nodes and in the Stack Node in order to produce the classification decision.

The method was evaluated on the ASVspoof 2019 dataset in the Logical Access track, which includes speech synthesis and voice conversion attacks, using the Equal Error Rate (EER) and minimum tandem Detection Cost Function (min t-DCF) metrics. Compared with the RawGAT-ST model, the AASIST system achieved an average EER of 1.13% compared with 1.39%, and an average min t-DCF of 0.0347 compared with 0.0443, a relative improvement of about 22% in the latter metric. In the best run AASIST achieved results of 0.83% EER and 0.0275 min t-DCF. In addition, a lightweight version called AASIST-L was presented, which includes only 85 thousand parameters and achieves an EER of 0.99% in the best reported run. Ablation experiments showed that the three architectural novelties contribute to the model's performance, with the removal of Heterogeneous Attention having the greatest negative effect.

Despite the strong results, the study was evaluated on ASVspoof 2019 LA only, and therefore does not provide direct evidence of the model's ability to generalize to later datasets or to forgeries based on Neural Audio Codecs. Likewise, although the MGO mechanism is intended to encourage the learning of complementary features, no analysis was presented that explicitly identifies which artifacts are learned in each of the model's branches. In addition, the differences between runs with different Random Seeds emphasize the importance of evaluating the model's stability alongside reporting its best results.

For the present study, AASIST constitutes a significant candidate for examining a Deep Learning-based approach that combines relations between temporal and spectral features. Accordingly, AASIST or its lightweight version AASIST-L can be compared with classical models based on MFCC and CQCC, and their generalization ability across different data sources and forgery methods can also be examined. This evaluation will be based on a controlled protocol, maintaining separation between the data sets and using EER as the central metric, while addressing performance stability and computational requirements.

Bibliographic source

Jung, J.-w., Heo, H.-S., Tak, H., Shim, H.-j., Chung, J. S., Lee, B.-J., Yu, H.-J., & Evans, N. (2022). AASIST: Audio Anti-Spoofing Using Integrated Spectro-Temporal Graph Attention Networks. *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 6367–6371. https://doi.org/10.1109/ICASSP43922.2022.9747766